

Recitation key

Question 1

Part (a): AD Shock from Optimism

When consumers and investors become more optimistic, the **aggregate demand (AD) curve shifts to the right**. In the quantity theory framework, this is equivalent to an increase in velocity $\vec{v}$, raising nominal spending growth ($\dot{M} + \vec{v}$).

- The new AD curve intersects the SRAS curve to the **right** of the original equilibrium.
- The intersection of the new AD with the SRAS ($E\pi = 10\%$) occurs at a **higher real GDP growth rate** (label: “**High-AD real growth**”) and a **higher inflation rate** (label: “**High-AD inflation**”).
- Since the SRAS is upward-sloping, both real growth and inflation rise above their initial values.

Key intuition: With expected inflation anchored at 10%, a positive demand shock moves the economy up and to the right along the SRAS, temporarily pushing real growth above potential and inflation above 10%.

Part (b): Central Banker’s Response

To **fully reverse** the AD shock “just right,” the central banker **tightens monetary policy** — reducing money growth $\dot{M}$ by exactly enough to offset the increase in velocity $\vec{v}$.

- The AD curve shifts **back to the left**, returning to its original position: $AD(\dot{M} + \vec{v} = 13\%)$.
- The economy returns to the **original equilibrium**: real GDP growth at the potential growth rate, and inflation at 10%.
- On the graph: draw the original AD curve again (or indicate the new AD shifts back to overlap with the original).

Part (c): Inflation Rate if Done “Just Right”

$$\pi = 10\%$$

If the central banker perfectly offsets the velocity shock by reducing $\dot{M}$ by the same amount that $\vec{v}$ increased, nominal spending growth remains at 13%. The economy stays at the intersection of the original AD curve and the LRAS, and inflation remains exactly **10%**.

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Question 2

Part (a): Lowering Inflation Without Reducing Real Growth

Using the quantity theory of money:

$$\dot{M} + \vec{v} = \pi + g_Y$$

With $\vec{v} = 0$ (no velocity shocks) and potential growth $g_Y^* = 3\%$, and the target inflation $\pi^* = 2\%$:

$$\dot{M} = \pi^* + g_Y^* - \vec{v} = 2\% + 3\% - 0\% = \boxed{5\%}$$

What the central banker should tell the public:

“We are committed to bringing inflation down to 2%. We will set money growth at 5% and maintain it. You should update your inflation expectations to 2%.”

Key mechanism: The central banker must **credibly commit** to the lower money growth rate so that the SRAS shifts down. If the public believes the announcement and lowers their expected inflation to 2%, the SRAS($E\pi = 10\%$) shifts down to SRAS($E\pi = 2\%$). The new equilibrium will be at:

$$\pi = 2\%, \quad g_Y = g_Y^* = 3\%$$

Real growth is preserved because the disinflation is **costless** when fully credible — the economy moves straight down along the LRAS rather than going through a costly recession.

Part (b): The Temptation to Inflate

Once the public has been convinced and has **lowered their inflation expectations to 2%**, the SRAS is now anchored at $E\pi = 2\%$. At this point, the central banker faces the following temptation:

- The central banker could **secretly increase money growth** above 5% (e.g., back to 13%), shifting the AD curve to the right.
- Because expectations are now anchored at 2%, the economy moves **up and to the right along** the new SRAS($E\pi = 2\%$).

- This produces a **boom in real GDP growth** above potential — which looks good in an election year.
- The cost (higher inflation) comes **later**, after the election, when expectations adjust upward.

Question 3

Part (a): How the Fed Cuts Rates from 2% to 1.5%

To lower the federal funds rate, the Fed will **increase the supply of reserves**.

- The reserve supply curve is **vertical** (perfectly inelastic) because the Federal Reserve is the **monopoly supplier** of reserves — it can set any quantity it chooses, independent of the interest rate.
- Reserve demand is downward-sloping: banks demand more reserves at lower interest rates (opportunity cost argument).
- To push the equilibrium rate **down** from 2% to 1.5%, the Fed shifts the vertical supply curve **to the right** by conducting **open market purchases** (buying Treasury securities and crediting banks with new reserves).
- The new intersection of the rightward-shifted vertical supply with the downward-sloping demand curve yields the lower equilibrium FFR of 1.5%.

Part (b): Effect on Bank Lending and the Money Supply

- When the FFR falls to 1.5%, the **return on holding reserves at the Fed decreases**.
- This lowers the **opportunity cost of lending**: banks now give up less by lending money out rather than holding reserves at the Fed.
- As a result, banks will **lend more money** — not merely because they have more reserves, but because holding reserves has become relatively less attractive.
- Through the **money multiplier**, this additional lending creates new deposits, which enables further lending.
- The net effect is an **increase in the nation's money supply**.

Part (c): Effect on Aggregate Demand

This will typically **increase aggregate demand**.

- A higher money supply raises nominal spending: from the quantity theory, $AD = \dot{M} + \vec{v}$, so an increase in $\dot{M}$ shifts the AD curve to the right.

- Lower interest rates also directly stimulate investment and consumption (the interest rate channel of monetary transmission).
- The rightward shift of AD raises both real GDP growth (in the short run) and inflation.